| Quantity | Value (as run) | Why this choice |
|---|---|---|
| **Method** | D2Q9, single-relaxation BGK | simplest collision that is adequate in the laminar regime |
| **Reynolds number** | 500 – 1000 | matches the pOnEllipse dataset; laminar shedding, St $\approx$ 0.2 |
| **Mach number** | $\approx$ 0.13 (held fixed) | weakly-compressible limit; error $O(\mathrm{Ma}^2) \approx$ 1.7% |
| **Relaxation $\tau$ / $\omega$** | 0.545 / 1.835 | fixed; $\tau - 0.5 = 0.045$ stability margin, $\omega < 2$ |
| **Resolution** | N $\approx$ 0.2·Re cells/chord (100 $\rightarrow$ 200) | scaled with Re to pin Ma $\approx$ 0.13 as $\nu$ drops |
| **Knudsen number** | $\approx 3 \times 10^{-4}$ | $\ll 1 \Rightarrow$ continuum / Navier-Stokes limit valid |
| **Wall model** | Bouzidi curved no-slip | sub-grid wall distance; no staircasing on arbitrary polygons |
| **Turbulence model** | none (DNS-resolved) | 2D laminar, Re $\leq$ 1000 — no LES needed |
| **Domain** | 24 chords (6·body·17) + sponge outlet | open free-stream box; sponge kills acoustic reflections |
| **Time-averaging** | $\geq$ 10 shedding periods after washout | unsteady shedding $\Rightarrow$ mean $C_d$, $C_l$, surface $C_p$ |
| **Cost (1 GPU, $\approx$ 1500 MLUPs)** | 60 s (Re500) $\rightarrow$ 7.5 min (Re1000) | per shape — cheap enough to sit in the design loop |
| **Validation** | surface $C_p$ vs dataset, $\rho$ = 0.97 – 0.999 | calibrates the solver as an independent ground truth |